The Mechanism of Fields: Supplementary Dynamical Proofs

Shao-Wu Hu
Shanghai, China

This supplement accompanies *The Mechanism of Fields from the Bost–Connes System*. Finite joint exchanges, matter carriers, and their identifications are inherited from Refs. [1, 2]. We give the actual operators, returns, and constraint proofs for the constructions in the main text. Preparation and spatial deployment are specified with each action. Energy refers to the actual self-adjoint generator of that construction; preparation and time-dependent control work enter the joint balance, and correspondence with bare-shell energy is checked for the exchange used. We use $\hbar = 1$ throughout.

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§D1 Loop Excitation and Energy Transfer

§D1.1 Source Spectrum and Actual Preparation

Take two edges of a triangle as a tree. Its matter action is

$$\begin{aligned} H_{\text{tree}} &= mY \otimes I_3 + Z \otimes T, \\ T &= \kappa(|1\rangle\langle 0| + |2\rangle\langle 1| + \text{h.c.}), \\ B &= \kappa Z \otimes |0\rangle\langle 2|, \end{aligned} \tag{D1.1}$$

Here $m, \kappa > 0$, $Y^2 = Z^2 = I$, and $\{Y, Z\} = 0$. The closing edge is $V = e^{i\Phi} B \otimes S_+ + e^{-i\Phi} B^\dagger \otimes S_-$. The shifts $S_\pm$ change the loop count n, which is neither position nor photon number. The complete finite Gauss sector is encoded by $|x, r_x + n\ell_C\rangle$, where r_x is tree-path flux and ℓ_C is triangle-loop flux. Retain complete tree units for $n = -M, \ldots, M-1$ and the endpoint $(x, n) = (0, M)$, with $M \geq 2$.

The tree position modes are $v_0 = (1, 0, -1)/\sqrt{2}$ and $v_\pm = (1, \pm\sqrt{2}, 1)/2$. With $Y = \sigma_z, Z = \sigma_x$, define

$$\begin{aligned} E &= \sqrt{m^2 + 2\kappa^2}, \\ \Delta &= E - m, \\ c &= \sqrt{\frac{1 + m/E}{2}}, \\ s &= \sqrt{\frac{1 - m/E}{2}}, \\ |E, \pm\rangle &= (c|+\rangle \pm s|-\rangle)v_\pm. \end{aligned} \tag{D1.2}$$

The lower positive branch is $g = |+\rangle v_0$, and the releasable upper combination is $b = (|E, +\rangle - |E, -\rangle)/\sqrt{2}$. Direct multiplication gives

$$\begin{aligned} P_g B P_E &= \beta |g\rangle\langle b|, \\ P_g B^\dagger P_E &= -\beta |g\rangle\langle b|, \\ \beta &= \kappa s/2, \\ P_g iq(B^\dagger - B)P_E &= -2iq\beta |g\rangle\langle b|. \end{aligned} \tag{D1.3}$$

The lowering block therefore belongs to the original current. The energy difference from b to g is Δ. The maintenance contribution increases by $m - m^2/E$, paid by a decrease $2\kappa^2/E$ in spatial motion, leaving exactly Δ.

The releasable preparation has an explicit process. In the basis $u = |+, 1\rangle$, $v = |-\rangle(|0\rangle + |2\rangle)/\sqrt{2}$, tree-edge phase control gives

$$
\begin{aligned}
H_{\mathrm{prep}} &= \begin{pmatrix} m & \sqrt{2}\kappa e^{-i\alpha} \\ \sqrt{2}\kappa e^{i\alpha} & -m \end{pmatrix}, \\
\vartheta &= \arcsin\sqrt{\frac{E}{2(E+m)}}, \\
\alpha &= \arccos\frac{m}{E+m}, \\
t_{\mathrm{prep}} &= \vartheta/E.
\end{aligned}
\tag{D1.4}
$$

Expanding the exponential using $H_{\mathrm{prep}}^2 = E^2 I$ gives $e^{-it_{\mathrm{prep}}H_{\mathrm{prep}}}u = e^{-i\psi}(cu+sv)$, with $\psi = \arctan[(m/E)\tan\vartheta]$. Restoring the original tree phase costs mean work $E-m$. The closing edge is off during preparation. Loading at the middle position, phase switching, and actual durations must be included. The upper-positive projection of a root-position input is orthogonal to b; this preparation cannot replace undriven evolution from that root input.

§D1.2 Complete Transport and Simultaneous Action

Define

$$
\begin{aligned}
\eta_x^{\Gamma} &= \frac{e^{i\Phi}|r_x + \ell_C\rangle - e^{-i\Phi}|r_x - \ell_C\rangle}{\sqrt{2}}, \\
\Gamma &= \frac{e^{i\Phi}|\ell_C\rangle - e^{-i\Phi}|-\ell_C\rangle}{\sqrt{2}}, \\
B_0 &= \sum_x |r_x, \Gamma_1\rangle\langle\eta_x^{\Gamma}, 0_1|.
\end{aligned}
\tag{D1.5}
$$

Input and output have the same divergence, and distinct x components are orthogonal. Thus B_0 is a Gauss-preserving partial isometry that retains all residual-source position coherence. Subsequent neutral cells use $B_r = |0_r, \Gamma_{r+1}\rangle\langle\Gamma_r, 0_{r+1}|$. With common routing $\nu = 2\kappa$ and independent light orientation $z = \pm 1$, define

$$
\begin{aligned}
H &= H_S + C + F, \\
H_S &= H_{\mathrm{tree}} + V, \\
C &= \frac{i\nu z}{2}(B_0 - B_0^{\dagger}), \\
F &= \frac{i\nu z}{2}\sum_{r=1}^{K-1}(B_r - B_r^{\dagger}).
\end{aligned}
\tag{D1.6}
$$

Take $K \geq 2$, with input $b, n = 0$ and all distant cells empty. Retain all 2^K distant occupation strings, both source frequency branches, and unreleased loops. Local matrix units give $[H_{\mathrm{tree}}, C] = 0$ and $[H_S, F] = 0$, hence

$$
\begin{aligned}
J_{\mathrm{in}} &= i[V, C], \\
J_{\mathrm{far}} &= i[C, F], \\
\langle \dot{H}_S \rangle &= -\langle J_{\mathrm{in}} \rangle, \\
\langle \dot{C} \rangle &= \langle J_{\mathrm{in}} - J_{\mathrm{far}} \rangle, \\
\langle \dot{F} \rangle &= \langle J_{\mathrm{far}} \rangle.
\end{aligned}
\tag{D1.7}
$$

Total energy is E. Let $j = z\nu/2$ and $A = \sqrt{2}\beta$. Actual action sequences give the amplitudes with residual source g and exactly one pattern in either the first or the second distant cell, with all other distant cells empty:

$$
\begin{aligned}
A_1(t) &= -ijAt^2/2 - jA(E + 2m)t^3/6 + O(t^4), \\
A_2(t) &= -ij^2At^3/6 - j^2A(E + 3m)t^4/24 + O(t^5).
\end{aligned}
\tag{D1.8}
$$

The distant-edge matrix element is ij. The fifth-order term in $2\,\mathrm{Re}(A_2^* ij A_1)$ vanishes; the sixth-order term is $j^4A^2(E - m)/72$. Other source branches, more distant positions, and multiple patterns do not contribute at this order. Therefore

$$
\langle F \rangle_t = \frac{\beta^2\nu^4\Delta}{576}t^6 + O(t^7) > 0 \quad \text{for sufficiently small } t > 0. \tag{D1.9}
$$

This proves positive net outward transfer in the specified finite process, not pure positive-frequency single-photon output. Switching off the contact and closing edge costs work $-\langle V + C \rangle$, which cannot be omitted. The next section gives the actual three-dimensional source and the scope of transverse preservation.

§D1.3 Three-Dimensional Local Excitation and Transverse Propagation

Take a periodic cubic lattice with at least three sites per axis. Let $T_i|x\rangle = |x + \ell e_i\rangle$, and use central differences with polar and axial vector components:

$$
\begin{gathered}
\delta_i = \frac{T_i^\dagger - T_i}{2\ell}, \qquad (\mathcal{C}w)_a = \sum_{ib} \epsilon_{aib}\delta_i w_b, \\
h = ic_\gamma \begin{pmatrix} 0 & \mathcal{C} \\ -\mathcal{C} & 0 \end{pmatrix}, \qquad c_\gamma = \nu\ell.
\end{gathered}
\tag{D1.10}
$$

The curl $\mathcal{C}$, and hence h, is self-adjoint, with each neighboring-edge strength equal to $\nu/2$. This central-difference deployment is distinct from the edge–face incidence of D2. Set $D_\perp = \mathrm{diag}(\mathrm{div}_\delta, \mathrm{div}_\delta)$; then $D_\perp h = 0$. Choose a local circulation direction a, and let R_Γ be the source-side loop retrieval block of B_0 in Eq. (D1.5). Define

$$
\begin{gathered}
w = \ell\mathcal{C}|0, a\rangle, \qquad f = \begin{pmatrix} w \\ 0 \end{pmatrix}, \qquad \|f\| = 1, \qquad D_\perp f = 0, \\
a_i^\dagger = |\Gamma_i\rangle\langle 0_i|, \quad a_f^\dagger = \sum_i f_i a_i^\dagger, \\
C_3 = ij(R_\Gamma \otimes a_f^\dagger - R_\Gamma^\dagger \otimes a_f), \qquad j = \nu/2.
\end{gathered}
\tag{D1.11}
$$

For $a = z$, the four nonzero position–component amplitudes of w are $\pm 1/2$, so normalization and vanishing divergence follow from the actual wiring. All distant occupation strings remain present. The operator $a_f^\dagger$ produces a normalized state on the vacuum input but need not be a partial isometry on general multiple-occupation states. Define $H_{\mathrm{far}} = \sum_{ij} h_{ij} a_i^\dagger a_j$ and $H_3 = H_{\mathrm{tree}} + V + C_3 + H_{\mathrm{far}}$. The original source identity $R_\Gamma V|b, 0\rangle = \sqrt{2}\beta|g, 0\rangle$ gives

$$P_{N_{\mathrm{far}} \geq 1} e^{-itH_3} \Psi_0 = -\frac{ij\sqrt{2}\beta}{2} t^2 |g, 0\rangle \otimes |f\rangle + O(t^3). \tag{D1.12}$$

The earliest distant output therefore enters the transverse space. The low-source channel containing either one source-side loop or one distant pattern exactly reduces $H_{0,3} = H_3 - V$. Let $|r_0\rangle$ denote the low source with one loop and distant vacuum. Its embedding J_g satisfies

$$H_{0,3} J_g = J_g (mI + h_{\mathrm{ext}}), \qquad h_{\mathrm{ext}} = \begin{pmatrix} 0 & -ij\langle f| \\ ij|f\rangle & h \end{pmatrix}. \tag{D1.13}$$

Extend $D_\perp$ by zero on the source slot to obtain $\widetilde{D}_\perp$; then $\widetilde{D}_\perp h_{\mathrm{ext}} = 0$. After removing the common m phase, the exact first-order excitation coefficient in a formal parameter λV is

$$\alpha^{(1)}(t) = -i\sqrt{2}\beta \int_0^t e^{-ih_{\mathrm{ext}}(t-s)} |r_0\rangle e^{-i\Delta s}\, ds, \qquad \widetilde{D}_\perp \alpha^{(1)}(t) = 0. \tag{D1.14}$$

After source closure, the vacuum single-pattern correlation is likewise given by $e^{-ith} f$. This linear channel retains source-side returns but does not replace complete source depletion and multiple-pattern evolution at $\lambda = 1$. The full hard-core space does not automatically obey the same closed linear transverse equation.

Since $[H_{\mathrm{tree}}, C_3] = 0$ and source and distant actions commute, the energy balance of Eq. (D1.7) remains valid with C_3, H_{far}. With $A = \sqrt{2}\beta$, the earliest net distant energy is

$$\langle H_{\mathrm{far}} \rangle_t = \frac{A^2 j^2 \Delta}{72} \langle f, h^2 f \rangle t^6 + O(t^7). \tag{D1.15}$$

This gives positive short-time net output for the stated nontrivial propagation deployment. It does not establish persistent one-way far-field radiation or remove the positive and negative frequencies and multiple patterns from the complete output.

§D2 Electromagnetic Returns, Finite Corrections, and Calibration

§D2.1 Return in an Isolated Spectral Band

For the finite flux and loop construction in the main text, use $[N_e, U_f] = \delta_{ef} U_f$, $[N_e, W_p] = C_{pe} W_p$, and $DC^\dagger = 0$. Let P project onto all auxiliaries in their upper inputs, with $E_R = \sum_r m_r$. If

$$|\delta| \|V\| + \delta^2 \|H_m\| < m_{\min}, \tag{D2.1}$$

the neighboring spectral clusters remain isolated. Identify the band through the continuous polar-decomposition isometry J_δ from P to the actual spectral band P_δ. Since $PVP = 0$, one action of V flips one auxiliary, and a second must flip the same auxiliary back to return to P. Thus

$$\begin{aligned} J_\delta^\dagger H(\delta) J_\delta &= E_R + \delta^2 (H_m + K_R) + O(\delta^4), \\ K_R &= PV(1-P)(E_R - H_R)^{-1}(1-P)VP. \end{aligned} \tag{D2.2}$$

Each edge contributes $\lambda_e^2 N_e^2/(2m_e)$, and each face contributes $\lambda_p^2 (I - W_p)^\dagger (I - W_p)/(2m_p)$. Common auxiliary parity changes δ to $-\delta$, so the band expansion is even. At fixed finite volume and nonzero gap, the remainder norm is bounded by $C\delta^4$. In slow time $\tau = \delta^2 t$, Duhamel's formula bounds the evolution error by $C\delta^2|\tau|$. Observables are transformed by the same J_δ; auxiliaries are not repeatedly projected. No uniform infinite-volume bound is asserted.

§D2.2 Finite Field Equations and Static Energy

Write $A_p = I - W_p$, $K_p = (W_p + W_p^\dagger)/2$, $S_p = (W_p - W_p^\dagger)/(2i)$, $\Pi_{\text{in}} = I - W_p^\dagger W_p$, and $\Pi_{\text{out}} = I - W_p W_p^\dagger$. Expanding both sides gives

$$A_p^\dagger A_p = S_p^2 + (I - K_p)^2 + \frac{1}{2}(\Pi_{\text{out}} - \Pi_{\text{in}}). \tag{D2.3}$$

Substitution in the return energy retains the nonlinear and cutoff terms beyond quadratic field energy. Taking commutators from $[N_e, W_p] = C_{pe} W_p$ yields

$$\dot{N} = -j/q - bC^\dagger S, \qquad \dot{S}_p\big|_{H_E} = \frac{a}{2} \sum_e C_{pe} \{N_e, K_p\}, \quad H_E = \frac{a}{2} \sum_e N_e^2. \tag{D2.4}$$

Adding $i[H_m + H_B, S_p]$ gives the finite Faraday correction in the main text. With D positive at the edge origin and negative at its end, the original matter current is $j = iq(T^\dagger - T)$, so $\dot{\rho} = -Dj$ is compatible with $DN = \rho/q$. The field electric-energy change caused by the current is $-a \sum_e \{N_e, j_e\}/(2q)$. Remaining magnetic boundary and contact terms are retained by the complete commutators.

On a connected zero-total-charge graph, $N_L = D^\dagger (DD^\dagger)^+ \rho/q$ is orthogonal to $N - N_L$, giving the longitudinal energy decomposition in the main text. In the noncompact localized-phase representation, $U_\rho = e^{iN_L \cdot \theta}$ translates flux and, because $CN_L = 0$, preserves all loop actions. On the reduced zero-charge space,

$$U_\rho^\dagger H_{\text{field}} U_\rho = H_{\text{field}} + \frac{a}{2}\|N_L\|^2. \tag{D2.5}$$

This proves the full spectral shift in that approximate representation. Finite integer flux generally disallows arbitrary real translation, so its lowest-energy differences must be computed in the actual sectors. The uniform cubic-complex symbol $d_i = e^{ik_i\ell} - 1$ gives spectrum $\{0, |d|^2, |d|^2\}$ for $C^\dagger C$, yielding the two transverse waves and long-wavelength speed.

§D2.3 Common Units and Relative Return

Fix $q_0 = q$ as the fundamental flux-charge unit of the main text. An actual transfer of charge $Q = nq_0$ carries $L_n = (U^\dagger)^n$ for $n \geq 0$ or $L_n = U^{-n}$ for $n < 0$, so $[N, L_n] = -nL_n$. The joint jump preserves $DN - \rho/q_0$ and gives current $j = iQ(T_n^\dagger - T_n)$. The integer-charge factor thus follows from the actual number of flux steps. Convert band frequency to actual reference time with common rate s and lattice spacing ℓ. The physical coefficients are

$$\varepsilon_{\text{phys}} = \frac{q_0^2}{sa\ell}, \quad \mu_{\text{phys}}^{-1} = sbq_0^2\ell, \quad c = s\ell\sqrt{ab}, \quad \frac{(nq_0)^2}{4\pi\varepsilon_{\text{phys}}c} = \frac{n^2}{4\pi}\sqrt{a/b}. \tag{D2.6}$$

For nonzero integer charge $n \neq 0$, matching an original coupling α_* requires $a/b = (4\pi\alpha_*/n^2)^2$; matching light speed separately constrains the product. A common rescaling $(a, b) = w(a_0, b_0)$, $w > 0$, leaves the dimensionless ratio unchanged. Only an independently justified relative return can change it. For the second-order action $Q_w \otimes H_f$ preserving each work branch, with initially independent work and field states, a mixture over w gives a mixture of conditional field evolutions, generally not the unitary evolution at mean w. This conclusion is not extended to arbitrary branch-mixing complete actions.

§D3 Massive Vector Propagation and Complete Endpoints

Express the original complete mass exchange B in the same time units as spatial propagation. Under continuous activity, its singular values directly give rest-frame half-gaps. For cycles in which processing preserves relative phase, while mass, transport, and endpoint contacts share an active interval, the activity fraction converts the entire generator [2]. Below, B, ν, the spectral parameter z, and the source and receiver couplings use this common time convention. Define

$$\mathcal{M} = \begin{pmatrix} 0 & B^\dagger \\ B & 0 \end{pmatrix}, \quad \Xi = \mathrm{diag}(I, -I), \quad \{\mathcal{M}, \Xi\} = 0. \tag{D3.1}$$

If $B^\dagger B u_j = M_j^2 u_j$, then $v_j = B u_j / M_j$ and $\phi_j^\pm = (u_j \oplus (\pm v_j))/\sqrt{2}$. On seven spatial branches, use Q, D_Q from the main text. Even for complex difference symbols, $[d]_\times^\dagger [d]_\times = |d|^2 I - d d^\dagger$, so $Q^\dagger Q = L I_3$. Cross terms vanish in the square of complete h. The invariant block between the three positive rest inputs and their negative-orientation transport outputs is

$$h_j(k) = \begin{pmatrix} M_j I_3 & \nu\sqrt{L(k)} I_3 \\ \nu\sqrt{L(k)} I_3 & -M_j I_3 \end{pmatrix}. \tag{D3.2}$$

Using $(z - h)^{-1} = (z + h)(z^2 - h^2)^{-1}$ and the zero compression of D_Q to the spatial inputs gives

$$R_j(z) = \frac{z + M_j}{z^2 - M_j^2 - \nu^2 L} I_3, \quad J_j^\dagger e^{-iht} J_j = \cos(\Omega_j t) - i \frac{M_j}{\Omega_j} \sin(\Omega_j t). \tag{D3.3}$$

If the endpoint couples only to u_j, its mass expectation is zero and the numerator becomes z. The mediator is gapped on the pure mass-bright space; unused mass-dark branches remain in the complete carrier.

Combine source and receiver columns into V. Block elimination gives $\Sigma(z) = V^\dagger (z - h)^{-1} V$, including self and cross returns. For $|z| < M_j$, the resolvent identity and gap give

$$\|\Sigma(z) - \Sigma(0)\| \le \frac{\|V\|^2 |z|}{M_j (M_j - |z|)}. \tag{D3.4}$$

In the same time unit, if $|z| \ll M_j$ and both endpoint profiles are supported where $\nu^2 L \ll M_j^2$, the local positive-rest cross coefficient is $-\lambda_R^* \lambda_S / M_j$. Otherwise the full spatial kernel remains. Without a separate normalization of the source fields, this coefficient cannot be relabeled as the Fermi constant. On N finite positions, the static propagation kernel is

$$G_j(r) = \frac{1}{N} \sum_k \frac{e^{ik \cdot r}}{M_j^2 + 4\nu^2 \sum_i \sin^2(k_i \ell / 2)}. \tag{D3.5}$$

For localized long-wavelength sources, $M_j / \nu \ll 1$, and positions far from boundaries, $G_j(r) \simeq \ell^3 e^{-M_j r / c} / (4\pi c^2 r)$, with $c = \nu \ell$. The factor ℓ^3 comes from lattice-source normalization. Actual forces and transitions still depend on the complete endpoint matrix elements.

§D4 Color Links and Noncommuting Returns

§D4.1 Finite Links and Complete Constraints

An explicit link carrier contains $|0\rangle, |ij\rangle, |\overline{ij}\rangle$, with $i, j = 1, 2, 3$, for a total of 19 states. Take

$$\begin{aligned} L^a &= 0 \oplus (-t^{a\mathsf{T}} \otimes I_3) \oplus (t^a \otimes I_3), \\ R^a &= 0 \oplus (I_3 \otimes t^a) \oplus (-I_3 \otimes t^{a\mathsf{T}}), \\ U_{ij}|0\rangle &= |ij\rangle/\sqrt{3}, \quad U_{ij}|\overline{kl}\rangle = \delta_{ik}\delta_{jl}|0\rangle/\sqrt{3}, \\ U_{ij}|kl\rangle &= \tfrac{1}{2}\sum_{mn} \epsilon_{mik}\epsilon_{njl}|\overline{mn}\rangle. \end{aligned} \tag{D4.1}$$

Direct action gives the left and right covariance relations of the main text and $\sum_a (L^a)^2 = \frac{4}{3}(0 \oplus I_9 \oplus I_9)$. In terms of normalized matrix elements $1, \sqrt{3}g_{ij}, \sqrt{3}g^*_{ij}$, let P_Λ project onto all matrix elements of a finite set of irreducible representations, and let M_g denote multiplication by the group matrix elements. The finite representation family $U_\Lambda = P_\Lambda M_g P_\Lambda$ has defect

$$I - \mathcal{U}_\Lambda^\dagger \mathcal{U}_\Lambda = P_\Lambda M_g^\dagger (I - P_\Lambda) M_g P_\Lambda \geq 0. \tag{D4.2}$$

This mathematical representation specifies finite matrices, rather than adding an infinite physical entity. The minimal 19-state defect is not small and does not establish a controlled continuum limit.

Matter and link Gauss generators are $G^a_x = Q^a_x + \sum_{x\to y} L^a_e + \sum_{y\to x} R^a_e$. An electric auxiliary exchanges its upper singlet with a lower adjoint octet through $T^E_e = \sum_a |a\rangle\langle s| \otimes L^a_e$. Lower-state generators are $(F^b)_{ca} = -if^{bca}$. Including them in the complete endpoint Gauss generator makes auxiliary and link changes cancel by covariance.

§D4.2 Matrix Gram Operator and Curvature

Take the orthonormal Hermitian matrix basis $T_0 = I/\sqrt{3}, T_a = \sqrt{2}t^a$. Let $A_p = I - W_p$ and $A^A_p = \mathrm{tr}(T_A A_p)$. The magnetic auxiliary's $1 \oplus 8$ outputs implement $T^B_p = \sum_A |A\rangle\langle s| \otimes A^A_p$, hence

$$\sum_A A^{A\dagger}_p A^A_p = \mathrm{tr}(A^\dagger_p A_p), \qquad H_{2,c} = H_{m,c} + \frac{\alpha}{2}\sum_{e,a}(L^a_e)^2 + \frac{\beta}{2}\sum_p \mathrm{tr}(A^\dagger_p A_p). \tag{D4.3}$$

The upper-band proof of D2 applies auxiliary by auxiliary. The band isometry can intertwine color, carrying the complete constraint into the band. Face exchanges are deployed simultaneously as finite joint actions. Replacing them by sequential edge gates requires accounting again for intermediate outputs and durations.

The small-loop expansion $W = I + i\ell^2 F + O(\ell^3)$ gives leading complete Gram term $\ell^4 \,\mathrm{tr}\, F^2$. Taking a trace before squaring misses this term because $\mathrm{tr}\, F = 0$. Substituting $\Pi^a_i = -L^a_e/\ell^2$ and $A_i = \theta_i/\ell$ gives continuum energy

$$H_c = H_{m,c} + \int d^3x \left[\frac{\alpha\ell}{2}\sum_{ia}(\Pi^a_i)^2 + \frac{\beta\ell}{4}\sum_{i<j,a}(F^a_{ij})^2\right]. \tag{D4.4}$$

Hamilton's equations in the small-phase conjugate window yield the coefficients κ_E, κ_B and covariant Gauss relation in the main text. The finite link still uses actual commutators. Near unitarity, representation tails, smoothness, and the time window must all be controlled. This proof does not establish continuum Yang–Mills theory for arbitrary cutoffs and scales.

§D5 Complete Nucleons and Pion Exchange

§D5.1 Nucleon Preservation and the Odd Spatial Vertex

Let J_N embed the complete four nucleon states of Ref. [2]. Fix $|p\uparrow\rangle = |\Phi_{p\uparrow}\rangle$ and $|n\uparrow\rangle = -|\Phi_{n\uparrow}\rangle$, and generate the normalized down-spin states with total spin lowering $\mathsf{S}_- = (\mathsf{S}_1 - i\mathsf{S}_2)/2$. The column order is $(p\uparrow, p\downarrow, n\uparrow, n\downarrow)$. The common minus sign on the neutron columns fixes the basis phase of the isospin Pauli representation. Here $\mathsf{S}_i = \sum_{j=1}^{3} \sigma_j^i$ and $\mathsf{T}_a = \sum_{j=1}^{3} \tau_j^a$ are the Pauli sums over the three constituent slots. Total spin and total isospin preserve this space, restricting to their Pauli representations. Thus

$$\mathsf{S}_i J_N = J_N(I \otimes \sigma_i), \quad \mathsf{T}_a J_N = J_N(\tau_a \otimes I), \quad \mathsf{S}_i \mathsf{T}_a J_N = J_N(\tau_a \otimes \sigma_i). \tag{D5.1}$$

Actual rotation generators are half these matrices. The deployed action is the collective product. The constituent sum $\sum_j \sigma_j^i \tau_j^a$ generally produces components outside the four-state space; its compression does not replace a preserving action. The original collective mode and color singlet remain part of the complete carrier. Collective spin–isospin action is the identity on these factors, while the corresponding spectral-port exchange acts as the identity on internal isospin–spin factors.

Write $\tau_\pm = (\tau_1 \pm i\tau_2)/2$. The pion isospin triplet is represented by $|\varphi_a\rangle = \sum_{rs} (\tau_a)_{rs} |r\bar{s}\rangle/\sqrt{2}$, with the original color and spin singlets. Their source column satisfies

$$\sum_a \tau_a |\varphi_a\rangle = \sqrt{2}\tau_- |\pi^+\rangle + \sqrt{2}\tau_+ |\pi^-\rangle + \tau_3 |\pi^0\rangle. \tag{D5.2}$$

Total nucleon–pion charge is conserved, as is total isospin for equal masses and coupling strengths of the triplet components. Between equal-parity nucleons and the pseudoscalar pion, use two spatial arms of equal strength and opposite phase:

$$V_x \chi = \frac{g_\pi}{2i\ell} \sum_{a,i} (\tau_a \otimes \sigma_i)\chi \otimes |\varphi_a, \chi_\pi^+; 0\rangle \otimes (|x + \ell e_i\rangle - |x - \ell e_i\rangle). \tag{D5.3}$$

Both arms enter the same record, and the actual interaction is $V_x + V_x^\dagger$. Fourier transformation gives $g_\pi \tau_a \boldsymbol{\sigma} \cdot \widehat{\boldsymbol{k}}$, with $\widehat{k}_i = \sin(k_i \ell)/\ell$. A permanent record distinguishing the arms destroys this cancellation. The dimension of g_π is frequency times length.

§D5.2 Scalar Propagation and Complete Return

Use the original pion mass exchange $\mathcal{M}_\pi$ and anticommuting orientation Ξ_π. Couple one scalar slot to three gradient auxiliaries:

$$Q = (d_1, d_2, d_3)^{\mathsf{T}}, \quad d_i = S_i - I, \quad h_\pi = \mathcal{M}_\pi \otimes I_4 + \nu \Xi_\pi \otimes \begin{pmatrix} 0 & Q^\dagger \\ Q & 0 \end{pmatrix}. \tag{D5.4}$$

On the accessible mass-bright branch $m > 0$, the positive rest scalar port gives

$$
\begin{aligned}
R_\pi(z) &= \frac{z+m}{z^2-m^2-\nu^2 L},\\
L &= Q^\dagger Q,\\
R_\pi(0) &= -mG,\\
G &= (m^2+\nu^2 L)^{-1}.
\end{aligned}
\tag{D5.5}
$$

The proof uses the same squared-action identity as D3. Gradient auxiliaries are not pion spin. If $R_x|x\rangle = |-x\rangle$, assign $-R_x$ to the scalar slot and $S_i R_x$ to directed-edge auxiliary i. Then $(S_i R_x) d_i (-R_x) = d_i$, so the complete propagation preserves parity.

Both nucleons remain present. Specify zero/one-pion capacity, with P the zero-pion block, Q_1 the same two nucleons with one pion, and $V = V_A + V_B$. Schur elimination gives

$$
\begin{aligned}
H &= \begin{pmatrix} H_P & V^\dagger \\ V & H_Q \end{pmatrix},\\
P(z-H)^{-1}P &= [z - H_P - \Sigma(z)]^{-1},\\
\Sigma(z) &= V^\dagger (z-H_Q)^{-1} V.
\end{aligned}
\tag{D5.6}
$$

For fixed nucleon positions and a common rest baseline subtracted from both blocks, $H_Q = I_{NN} \otimes h_\pi$. On a uniform inversion-symmetric graph, define

$$
K_{ij}(r) = \frac{1}{N}\sum_k \frac{e^{ik\cdot r}\widehat{k}_i \widehat{k}_j}{m^2 + 4\nu^2 \sum_l \sin^2(k_l \ell/2)}.
\tag{D5.7}
$$

Take the same real strength g_π at both endpoints. Substitution of the actual source columns yields

$$
\begin{aligned}
\Sigma_{AB}(0) + \Sigma_{BA}(0) &= -2g_\pi^2 m (\boldsymbol{\tau}_A \cdot \boldsymbol{\tau}_B) \sum_{ij} \sigma_A^i \sigma_B^j K_{ij}(r),\\
\Sigma_{AA}(0) &= -3 g_\pi^2 m \,\mathrm{tr}\, K(0) I_{16}.
\end{aligned}
\tag{D5.8}
$$

The second identity uses the real symmetric coincident kernel; the receiver self return is analogous. The two directions give factor two, and the three isospin components are already summed in $\boldsymbol{\tau}_A \cdot \boldsymbol{\tau}_B$. For different phases or nonuniform space, retain $V_A^\dagger R_\pi V_B$ and its adjoint directly.

§D5.3 Continuum Kernel and Low-Energy Scope

For $m/\nu \ll 1$, long wavelengths, and points far from boundaries, set $c = \nu\ell$ and $\kappa = m/c$. Then

$$
\begin{aligned}
K_{ij}(r) &\simeq -\ell^3 \partial_i \partial_j Y(r),\\
Y(r) &= \frac{e^{-\kappa r}}{4\pi c^2 r},\\
g_c &= g_\pi \ell^{3/2},\\
V_{AB} &= 2 g_c^2 m (\boldsymbol{\tau}_A \cdot \boldsymbol{\tau}_B) \sigma_A^i \sigma_B^j \partial_i \partial_j Y.
\end{aligned}
\tag{D5.9}
$$

A radial function obeys $\partial_i\partial_j Y = (Y'' - Y'/r)\widehat{r}_i\widehat{r}_j + (Y'/r)\delta_{ij}$. Substitution of $Y'/Y = -\kappa - 1/r$ and $Y''/Y = \kappa^2 + 2\kappa/r + 2/r^2$ gives the central and tensor decomposition in the main text. As a distribution, it also contains

$$V_{\text{contact}} = -\frac{2g_c^2 m}{3c^2}(\boldsymbol{\tau}_A \cdot \boldsymbol{\tau}_B)(\boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B)\delta^{(3)}(r). \tag{D5.10}$$

Use the finite kernel or actual source shape at short distance. Identical nucleons require odd $L + S + I$. The resolvent identity of D3 gives, for $|z| < m$,

$$\|\Sigma(z) - \Sigma(0)\| \leq \frac{\|V\|^2|z|}{m(m - |z|)}. \tag{D5.11}$$

Using static return as a low-energy action additionally requires $\|V\|/m \ll 1$ and slow nucleon motion. Actual recoil enters nucleon motion in both blocks; force follows from the momentum commutator of the same effective action. Multipion processes and nuclear bound-state spectra are not determined by this capacity construction.

§D6 Finite-Time Control of Common Clocks and Rods

Matter maintenance, motion, and reception act on the same complete carrier. Take a matter carrier for which Ref. [2] establishes maintenance and position intertwining, and denote its isometric embedding by J, $\widehat{K}$ the actual maintenance and motion generator, and $\widehat{P}_x$ the physical position projector. The loading used retains matter position, and position-controlled excursions and returns preserve the corresponding blocks. The intertwining relations are

$$\widehat{K}J = JK, \qquad \widehat{P}_x J = JP_x, \qquad J^\dagger J = I.$$

For carriers whose position loading includes accompanying resources, physical translation satisfies $(S_X S_R)J = JS_X$ [2]. Using this translation in $\widehat{K}$ preserves the relative position of matter and its contact resources. Self-adjointness implies that $JJ^\dagger$ commutes with $\widehat{K}$; repeated multiplication and the finite spectral expansion then give $F(\widehat{K})J = JF(K)$. Readout by a fresh pointer w therefore satisfies

$$e^{-it\widehat{K}\otimes Y_w}(J \otimes I_w) = (J \otimes I_w)e^{-itK\otimes Y_w}.$$

This relation applies to arbitrary inputs in the code subspace, including correlations with old records, and carries each pointer output separately. The finite-angle writing in D7 thus uses the same source spectrum.

Reception additionally uses work package B and background packages V_x, on degrees of freedom distinct from the accompanying resources within J. Let S_x exchange the two new packages, $\widehat{C} = \sum_x \widehat{P}_x \otimes S_x$, $C = \sum_x P_x \otimes S_x$, and $\widetilde{J} = J \otimes I_{\text{env}}$. The actual reception $\widehat{H}_C = \widehat{C}^\dagger(\widehat{K} \otimes R_B)\widehat{C}$ satisfies

$$\widehat{C}\widetilde{J} = \widetilde{J}C, \qquad \widehat{H}_C\widetilde{J} = \widetilde{J}H_C,$$
$$\tfrac{1}{2}\{\widehat{P}_x, \widehat{K}\}J = J\tfrac{1}{2}\{P_x, K\},$$

Here $H_C = C^\dagger(K \otimes R_B)C$. The first line follows by multiplying the position intertwining and $\widehat{K}J = JK$ term by term; expanding both anticommutators gives the second. These relations carry package reception in this section and local source action in D7. We use K, P_x to calculate these contacts below, retaining the source, work resources, and old records within the carrier. Charged or flavor-changing contacts are calculated separately using the ports and complete matrices specified in D1–D5.

§D6.1 Single-Package Response and Three-Package Overlap

In this matter representation, let K contain maintenance and position action. Let S_x exchange work unit B with the background unit at x. For $0 \le R \le I$,

$$C = \sum_x P_x \otimes S_x, \quad H_C = C^\dagger(K \otimes R_B)C, \quad (H_C)_{yx} = K_{yx} \otimes S_y R_B S_x. \tag{D6.1}$$

The complete matter input is independent of the new environment; correlations within matter and with its old records may remain. The new environment is $\eta = \sigma_B \otimes \prod_x \sigma_x$. Contracting the swap indices gives

$$\begin{aligned} \bar{H}_{xx} &= K_{xx} a_x, \\ a_x &= \mathrm{tr}(R\sigma_x), \\ \bar{H}_{yx} &= K_{yx} t_{yx}, \\ t_{yx} &= \mathrm{tr}(R\sigma_x \sigma_B \sigma_y) \\ (x &\neq y). \end{aligned} \tag{D6.2}$$

The reverse edge is the complex conjugate, ensuring self-adjointness. The coefficient t is an amplitude, not a probability. The actual return fixes the order of noncommuting factors. Each σ must describe the entire exchanged unit, including records or addresses if these are exchanged with it.

§D6.2 Error Bound for Finite Arrays

Deploy M actual independent arrays, each simultaneously coupled to the same matter with strength $1/M$:

$$\begin{aligned} H_M &= M^{-1} \sum_{a=1}^{M} H_a, \\ \Omega_0 &= \rho_T \otimes \eta^{\otimes M}, \\ V_\eta &= \mathbb{E}(H_a^2) - \bar{H}^2 \geq 0, \\ v^2 &= \|V_\eta\|. \end{aligned} \tag{D6.3}$$

The conditional expectation $\mathbb{E}$ averages only the input environment. For $\Delta_a = H_a - \bar{H}$, independence gives $\mathbb{E}(\Delta_a \Delta_b) = 0$ for $a \neq b$, entry by entry in the system matrix. This does not require the H_a, which share matter, to commute. Consequently,

$$\mathbb{E}[(H_M - \bar{H})^2] = V_\eta / M, \qquad v \leq \|K\|. \tag{D6.4}$$

Fix a purification of the input environment. The difference action on the comparison trajectory $e^{-is\bar{H}}\psi \otimes \chi^{\otimes M}$ has norm at most $v/\sqrt{M}$. Duhamel's identity and contraction under partial trace give

$$\begin{aligned} \|(e^{-itH_M} - e^{-it\bar{H}} \otimes I)\Psi_0\| &\leq |t| v/\sqrt{M}, \\ \|\rho_T(t) - e^{-it\bar{H}} \rho_T e^{it\bar{H}}\|_1 &\leq \min\{2, 2|t|v/\sqrt{M}\}. \end{aligned} \tag{D6.5}$$

An inert purification covers mixed states and spectator systems. This bound retains actual backreaction and subsequent correlations without resetting the environment. If the normalized reference coherence has unit modulus, and every unit real linear combination of its two orthogonal self-adjoint quadrature readouts has norm at most one, then $\varepsilon = 2|t|v/\sqrt{M} < 1$ bounds the modulus below by $1 - \varepsilon$ and the accumulated phase error above by $\arcsin \varepsilon$. The output bound does not directly control the instantaneous phase derivative.

§D6.3 Clock and Spatial Principal Part for an Explicit Preparation

Fix an original source eigenvalue λ_s. After reading area θ_x, retain orthogonal pointer records and exchange only the occupation unit:

$$\begin{aligned} h_x &= \frac{1-\sin(2\theta_x\lambda_s)}{2}, \\ \sigma_x &= \operatorname{diag}(h_x, 1-h_x), \\ \sigma_B &= \operatorname{diag}(h_B, 1-h_B), \\ R &= |0\rangle\langle 0|. \end{aligned} \tag{D6.6}$$

Substitution in D6.2 gives $a_x = h_x$ and $t_{yx} = h_B h_x h_y$. For the original three-direction matter action $K = mY + \nu \sum_i \alpha_i D_i$, the mean action is

$$\bar{H} = mYh + \nu h_B \sum_i \alpha_i h D_i h. \tag{D6.7}$$

In a uniform background, $E^2 = m^2h^2 + \nu^2 h_B^2 h^4 \sum_i \widehat{k}_i^2$. For $h, h_0, h_B > 0$, keep the same activity fraction in the field and reference preparations, and calibrate time and speed against the same source-free h_0 and work preparation h_B. The rest-clock and spatial-propagation factors are

$$A = h/h_0, \quad C_m = h^2/h_0^2 = A^2, \qquad B = A/C_m = 1/A. \tag{D6.8}$$

Here B is the spatial length-scale factor. For cycles in which maintenance and propagation share an active interval and processing preserves relative phase, a change in activity fraction by a ratio r_f gives $A = r_f h/h_0$, $C_m = r_f h^2/h_0^2$, and hence $B = r_f/A$; loading, readout, and retrieval repeated each cycle contribute to this ratio. The uniform result is exact for the comparison generator. Slowly varying backgrounds additionally require long-wavelength and gradient control; D6.5 controls actual finite-time outputs. Light acted on by the same type of reception has the same spatial factor after actual common deployment and baseline speed calibration. Static nonuniform light action $hK_\gamma h$ preserves the weighted condition $\operatorname{Div}(h^{-1}\psi) = 0$. Time-dependent h produces an additional weight derivative, so the static constraint does not carry over directly.

This construction has different inputs and service from the original complete coherent-package construction with a common multiplier. It proves a conditional reciprocal clock–rod relation, not that every background satisfies D6.8. Independent preparation, actual capacity, simultaneous coupling, and loading and record costs in a common time remain part of its scope.

§D7 Persistent Source Feedback and Local Light Constraints

§D7.1 Source Writing and Joint Energy

Couple the actual source generator K_s to a ready $|+x\rangle$ pointer. For the matter source carried through D6, K_s is the maintenance and motion part, with the original background reference retained on its own degrees of freedom. With unavailable occupation $N = (I - Z)/2$,

$$U_w = e^{-i\theta K_s \otimes Y}, \qquad \langle N \rangle - \tfrac{1}{2} = \tfrac{1}{2} \operatorname{tr}[\rho_s \sin(2\theta K_s)]. \tag{D7.1}$$

In the source eigenbasis, reduced coherence is multiplied by $\cos[\theta(\epsilon_a - \epsilon_b)]$. The same process thus produces background writing and source-coherence backreaction. Isolated writing commutes with K_s and does not require source energy to fall.

Finite-angle readout depends on the entire action actually coupled. If the contact includes an additional reference term B_{ref}, the output becomes $\frac{1}{2} \operatorname{tr} \rho \sin[2\theta(K_s + B_{\mathrm{ref}})]$. The condition $\langle B_{\mathrm{ref}} \rangle = 0$ alone does not remove it: for $K_s = E$ and an independent reference taking $\pm b_0$ with equal weights, the output is $\frac{1}{2} \sin(2\theta E) \cos(2\theta b_0)$. D7.1 uses the matter–motion action actually coupled; any reference compensation includes its own action and elapsed duration.

Divide each position edge equally between its endpoint pointers. The actual self-adjoint source and complete action are

$$\begin{aligned} J_x &= \tfrac{1}{2}\{P_x, K_s\}, \\ \sum_x J_x &= K_s, \\ V_w &= g \sum_x J_x \otimes Y_{w_x}, \\ H_{\mathrm{tot}} &= K_s + H_{\mathrm{ref}} + H_B + V_w + H_{\mathrm{tr}} + H_C. \end{aligned} \tag{D7.2}$$

Here H_{tr} exchanges complete packages, N_{all} sums occupations of the new pointers, work packages, and network packages, and $H_B = \omega N_{\mathrm{all}}$ is the energy of this new background network. The reception H_C uses $R_B = I - N_B$ from D6.3. The original reference action still participating within the matter carrier is denoted by H_{ref} and accounted for separately from the new network; take $H_{\mathrm{ref}} = 0$ when it is absent during this interval. All terms except V_w preserve this total occupation. Since $i[Y, N] = X$,

$$\frac{d}{dt}\langle N_{\mathrm{all}} \rangle = g \sum_x \langle J_x X_{w_x} \rangle, \qquad \dot{E}_a = \sum_b i\langle [H_b, H_a] \rangle, \qquad \sum_a \dot{E}_a = 0. \tag{D7.3}$$

Independent ready inputs give the initial rate of change of occupation $g\langle K_s \rangle$; subsequent rates retain source–pointer correlations. Each pair of energy currents is antisymmetric, balancing source, original reference, new background, matter, and contact energy at the same time. Preparation and switching work is separately $\int \langle \partial_t H \rangle dt$.

§D7.2 Evolving Reference and Correlation Corrections

Let T denote the receiving target and E all other actual participants. Use input $\rho_T \otimes \eta_0$ and define $\mathbb{E}_t(O) = \mathrm{Tr}_E[O(I_T \otimes \eta_{\mathrm{ref}}(t))]$. Set

$$\begin{aligned} H_{\mathrm{tot}} &= I_T \otimes H_E + H_M, \\ \eta_{\mathrm{ref}}(t) &= e^{-itH_E}\eta_0 e^{itH_E}, \\ \bar{H}(t) &= \mathbb{E}_t(H_M), \\ V(t) &= \mathbb{E}_t(H_M^2) - \bar{H}(t)^2 \geq 0. \end{aligned} \tag{D7.4}$$

The reference retains source and internal environmental correlations, separating only target reception. It is not the actual environmental marginal of the full state. With one fixed purification, compare complete evolution to $U_{\bar{H}}(t) \otimes U_E(t)$. The squared norm of the difference action on the comparison trajectory is $\langle V(t) \rangle$. The same Duhamel argument yields

$$\begin{aligned} \|\rho_T(T) - U_{\bar{H}}(T)\rho_T(0)U_{\bar{H}}(T)^\dagger\|_1 &\leq \min\left\{2,\ 2\int_0^T \sqrt{\|V(t)\|}\, dt\right\}, \\ i\dot{U}_{\bar{H}} &= \bar{H}(t)U_{\bar{H}}. \end{aligned} \tag{D7.5}$$

For initial target–environment correlations, define $\epsilon_0 = \|\Omega_{\mathrm{actual}}(0) - \rho_T \otimes \eta_0\|_1$ and add ϵ_0 to the right-hand side of Eq. (D7.5), with the total bound still capped at two. This is the full trace norm, without the factor $1/2$ used in trace distance. Under a shared source, define $\Delta_a(t) = H_{C,a} - \mathbb{E}_t(H_{C,a}) \otimes I_E$. The array variance is $M^{-2}\sum_{a,b}\mathbb{E}_t(\Delta_a\Delta_b)$. Cross covariances generally remain, so the $1/\sqrt{M}$ bound of D6 does not follow automatically.

If the work package remains independently diagonal in the reference, $\sigma_B = \mathrm{diag}(h_B, 1-h_B)$, the two network packages may still be arbitrarily correlated. The complete swap trace gives

$$\begin{aligned} t_{yx} &= h_B P_{xy}(00) + (1-h_B)(\eta_{xy})_{01,10}, \\ t_{yx} - h_B h_x h_y &= h_B\,\mathrm{Cov}(R_x, R_y) + (1-h_B)(\eta_{xy})_{01,10}. \end{aligned} \tag{D7.6}$$

For example, initially independent diagonal occupations a, b transported by $e^{-iu\mathrm{SWAP}}$, with $c = \cos u, s = \sin u$, give $h_x = c^2a + s^2b$, $h_y = c^2b + s^2a$, $P(00) = ab$, and $\eta_{01,10} = ics(a-b)$. Transport itself therefore makes single-package occupation insufficient to determine spatial response.

With field-free calibrated mass m and reference speed $c_0 = 1$, the long-wavelength energy of the D6 preparation is $E(A,p) = \sqrt{m^2A^2 + A^4p^2}$. At the field-free baseline,

$$\partial_A E|_1 = \frac{m^2 + 2p^2}{\sqrt{m^2+p^2}} = E + \boldsymbol{p}\cdot\boldsymbol{v}. \tag{D7.7}$$

The actual conjugate variable must be defined on the carrier retaining the record. Take the accessible occupation–record preparation and action

$$
\begin{aligned}
|\chi(\vartheta_x)\rangle &= \cos\vartheta_x|00\rangle + \sin\vartheta_x|11\rangle,\\
h_x &= \cos^2\vartheta_x,\\
T_x &= -i|00\rangle\langle 11| + i|11\rangle\langle 00|.
\end{aligned}
\tag{D7.8}
$$

The operator T_x is zero on the other two states. Differentiating $\chi(\vartheta) = e^{-i\vartheta T}|00\rangle$ in the preparation angle, with fixed input matter and work package, gives

$$
\begin{aligned}
i\langle[H_C, T_x]\rangle_0 &= -\partial_{\vartheta_x}\langle H_C\rangle_0 = \sin(2\vartheta_x)\langle\partial_{h_x}\bar{H}\rangle_0,\\
i\Big\langle[H_M, \sum_{a,x} T_{a,x}]\Big\rangle_0 &= 2\tan\vartheta\,\partial_{\log A}E.
\end{aligned}
\tag{D7.9}
$$

The last identity uses uniform preparation, a positive-energy eigenstate in the selected momentum block, and all arrays; their $1/M$ couplings cancel in the sum. Equation D7.7 is the energy-variation coefficient, while the actual torque also carries the preparation factor. Inaccessible archived records do not automatically supply an operational readout. Persistent evolution also contributes $i[H_E, T_x]$. Equation D7.7 cannot replace the net writing in D7.3. Scalar closure, a long-term working window, and an exponential relation do not follow from this identity.

§D7.3 Constraints on Three-Dimensional Local Light Sources

Let K_γ be a three-dimensional transverse propagation satisfying $\operatorname{Div} K_\gamma = 0$. On a transverse input, the local symmetric source gives

$$
\operatorname{Div}\big(\tfrac{1}{2}\{P_x, K_\gamma\}\psi\big) = \tfrac{1}{2}\operatorname{Div}(P_x K_\gamma \psi),
\tag{D7.10}
$$

The right-hand side is generally nonzero because position projection does not commute with divergence. Constraint preservation in common global writing or a one-dimensional route does not remove this term. Choose a fixed positive energy scale $\Lambda > 0$, independent of the input energy. The two-sided action $J'_x = K_\gamma P_x K_\gamma/\Lambda$ preserves transversality, but

$$
\sum_x J'_x = K_\gamma^2/\Lambda, \qquad \langle\psi_E, \sum_x J'_x\psi_E\rangle = E^2/\Lambda.
\tag{D7.11}
$$

This changes the total source spectrum and cannot simultaneously be called the original E writing. A joint realization of the complete local source, directional constraint, and persistent service is still required. The result excludes direct identification of these two simple alternatives.

§D8 Directional Radiation, Free Spectrum, and Reception

§D8.1 Complete Carrier and Invariant Constraints

Use six components of $\mathrm{Sym}^2(\mathbb{C}^3)$, two propagation arms, and a distinct unexcited state as the background carrier. For centered-difference symbol $d_i = \sin(k_i\ell)/\ell$, define

$$C = i[d]_\times, \quad r = |d|, \quad \mathcal{L}T = (CT + TC^{\mathsf{T}})/2, \quad h_g = ic_g \begin{pmatrix} 0 & \mathcal{L} \\ -\mathcal{L} & 0 \end{pmatrix}. \tag{D8.1}$$

Both C and $\mathcal{L}$ are self-adjoint, hence so is h_g. Symmetry gives $\operatorname{tr} \mathcal{L}T = 0$. With $\operatorname{Div} T = Td$, one has $\operatorname{Div} \mathcal{L}T = (C/2) \operatorname{Div} T$. Free propagation therefore preserves transverse traceless constraints.

The spectrum of C is $0, \pm r$, and its half-sum spectrum on symmetric tensors is

$$\operatorname{spec} \mathcal{L} = \{-r, -r/2, 0, 0, r/2, r\}. \tag{D8.2}$$

The complete two-arm spectrum has $\pm c_g r$, each twice, $\pm c_g r/2$, each twice, and four zero modes. At $r \neq 0$, the transverse traceless part contains only the first pair and has two positive-frequency modes. All remaining modes stay in the complete carrier. Invariance is not dynamical attraction, and negative frequencies are not removed.

Let $M = C^2$. The actual three-difference source is

$$\mathcal{B}_3(J) = \mathcal{L}(CJC^{\mathsf{T}}) = \tfrac{1}{2}(CJM - MJC) = \tfrac{1}{3}(4\mathcal{L}^3 - r^2\mathcal{L})J, \quad \mathcal{B}_3^2 = r^6 P_{\mathrm{TT}}. \tag{D8.3}$$

Transversality of the first expression follows from $Cd = 0$, and its trace vanishes by cyclicity. The last two identities follow on each eigenspace in D8.2. The projector P_{TT} expresses a spectral identity and is not an actual output filter. At $r = 0$, the source vanishes; dark directions remain.

§D8.2 Complete Source and Two-End Response

Use the original finite moving matter $K_0 = \mu B + c_s \boldsymbol{\alpha} \cdot \boldsymbol{p}$, with branches $P_\pm$, $E = \sqrt{\mu^2 + c_s^2 p^2}$, and $\Delta = 2E$. Within a specified fixed-momentum spectral block, the directional action and selected lowering blocks are

$$\begin{aligned} J_{ij} &= \frac{c_s}{2}(\alpha_i p_j + \alpha_j p_i), \\ S_a &= P_- J_a P_+, \\ f_{x,a} &= (0, \ell^3 \mathcal{B}_3 |x, E_a\rangle), \\ W &= \sum_a S_a \otimes f_{x,a}. \end{aligned} \tag{D8.4}$$

The E_a are six orthonormal symmetric basis tensors, and $J_a = \sum_{ij} (E_a)_{ij} J_{ij}$. This construction separately specifies an energy-resolved emission–absorption port

that preserves total count. The bare action J_a does not automatically contain only the lowering block, and this selection is not a universal local emission law for arbitrary moving matter. The interaction is $g\sum_a[S_a \otimes b^\dagger(f_{x,a}) + \text{h.c.}]$, while free field action preserves excitation number. Lowering emission and adjoint absorption preserve total count $P_+ + N_f$. Starting from the upper matter branch and unexcited field, the count-one space is exactly invariant. Subtracting the common lower-branch energy $-E$ gives

$$H_1 = \begin{pmatrix} \Delta I & gW^\dagger \\ gW & H_F \end{pmatrix}, \qquad H_F = I \otimes h_g. \tag{D8.5}$$

Every f has zero divergence and trace. Extending these constraints to the direct sum, the free intertwining identity in D8.1 and $\mathcal{G}W = 0$ give $\widetilde{\mathcal{G}}H_1 = h_c\widetilde{\mathcal{G}}$. Thus the complete source and return preserve the selected constraints.

Add a complete receiver of the same type, retaining both bodies' upper and lower branches and internal degeneracies. After subtracting the common lower-branch energy of both bodies, the count-one action becomes

$$H_1 = \begin{pmatrix} \Delta_S I & 0 & g_S W_S^\dagger \\ 0 & \Delta_D I & g_D W_D^\dagger \\ g_S W_S & g_D W_D & H_F \end{pmatrix}. \tag{D8.6}$$

The constraint proof continues term by term. Eliminating the field gives cross response $\Sigma_{DS} = g_D g_S W_D^\dagger (z - H_F)^{-1} W_S$. Factoring out actual material matrices and position phases, the axial-arm kernel is

$$\ell^6 \mathcal{B}_3^\dagger [(z - h_g)^{-1}]_{vv} \mathcal{B}_3 = \frac{\ell^6 z r^6}{z^2 - c_g^2 r^2} P_{\mathrm{TT}} \quad (r \neq 0). \tag{D8.7}$$

The proof uses $[(z - h_g)^{-1}]_{vv} = z(z^2 - c_g^2\mathcal{L}^2)^{-1}$ and D8.3. Neither endpoint's r^3 factor can be removed. Continuing the compressed accessible TT kernel to $z \to 0$ gives zero same-arm cross response; it cannot serve as the original static gravity kernel.

For a pure source input, the single-source field energy expands as

$$\langle H_F \rangle_t = \frac{g^2 \Delta t^4}{6} \langle W^\dagger H_F^2 W \rangle + O(t^5). \tag{D8.8}$$

The lowest t^2 term contains $W^\dagger H_F W = 0$, because the source enters one arm and H_F exchanges arms. The t^3 term also vanishes. Expansion of the complete exponential to fourth order gives D8.8. The coefficient is positive when $g \neq 0$ and $H_F W u_0 \neq 0$, but H_F retains both spectral signs. Positive short-time net output does not establish positive-definite macroscopic radiation energy or persistent far-field power.

From the excited-source input, the two-body action gives

$$\begin{aligned} d(t) &= -\tfrac{1}{2} g_D g_S t^2 W_D^\dagger W_S u_0 + O(t^3), \\ P_D(t) &= \tfrac{1}{4} g_D^2 g_S^2 t^4 \| W_D^\dagger W_S u_0 \|^2 + O(t^5), \\ E_S + E_D + E_F + E_{\mathrm{int}} &= E_{\mathrm{initial}}. \end{aligned} \tag{D8.9}$$

Receiver energy is $E_D = \Delta_D \|d(t)\|^2$. If local supports do not overlap, the displayed leading coefficient vanishes and actual arrival is determined by propagation

paths. For an existing incident axial tensor wave packet ϕ_v, set $T' = [\ell^3 \mathcal{B}_3 \phi_v](x_D)$, the filtered local tensor at receiver position x_D. At fixed wave vector, this may be expressed through the tensor amplitude at that position. With an equally weighted lower-branch receiver preparation, the absorption coefficient is

$$R_D(\phi_v) = c_D^2 \left[\|T' \boldsymbol{p}_D\|^2 - \frac{c_D^2}{(E_D^{(0)})^2} |\boldsymbol{p}_D^{\mathsf{T}} T' \boldsymbol{p}_D|^2 \right]. \tag{D8.10}$$

Here $E_D^{(0)} = \sqrt{\mu_D^2 + c_D^2 p_D^2}$ and $\Delta_D = 2E_D^{(0)}$. The coefficient follows from the Clifford trace of S_a, with actual short-time probability $g_D^2 t^2 R_D + O(t^3)$. Its second term retains polarization information. This finite motion endpoint is dark at $p_D = 0$. Internal material motion, permanent records, and resources must be supplied by the actual detector process.

§D8.3 Relation to the Original Material Readout

The original single-index direction operator is $Q_{ij} = (|i\rangle\langle j| + |j\rangle\langle i|)/2 - \delta_{ij} I/3$. Its natural two-index lift is

$$\widetilde{Q}_{ij} T = \tfrac{1}{2}(Q_{ij} T + T Q_{ij}^{\mathsf{T}}), \qquad \widetilde{q}_{ij}(T) = \mathrm{Re}(T T^{\dagger})_{ij} - \tfrac{1}{3}\delta_{ij} \|T\|^2. \tag{D8.11}$$

At fixed nonzero wave vector, any transverse traceless $T = aE_+ + bE_\times$ gives $\widetilde{q} = W(I/6 - nn^{\mathsf{T}}/2)$, where $W = |a|^2 + |b|^2$. Equally intense linear polarizations have the same original readout. General wave packets retain interference between distinct wave vectors and do not obey a single-direction formula.

The actual contact also satisfies $\mathrm{tr}(\widetilde{Q}_{ij} T) = T_{ij}$ for $\mathrm{tr}\, T = 0$, generally producing a trace. Some matrix elements also produce longitudinal components. Directly adding the original material contact to D8.6 therefore invalidates the preceding constraint theorem.

If coherence is separately prepared between the zero-frequency trace excitation $T_0 = I/\sqrt{3}$ and a TT amplitude, let real η define $T = (T_0 + \eta \delta T)/\sqrt{1 + \eta^2 \|\delta T\|^2}$. Then

$$\widetilde{q}(T) = \frac{(2\eta/\sqrt{3})\, \mathrm{Re}\, \delta T + \eta^2 [\mathrm{Re}(\delta T \delta T^{\dagger}) - \|\delta T\|^2 I/3]}{1 + \eta^2 \|\delta T\|^2}. \tag{D8.12}$$

A coherent reference thus permits a first-order directional readout. The state T_0 is an actual trace-direction excitation, not the unexcited state; the original pure-source, unexcited-field input does not contain it. Formation and preservation of the reference, its parent energy, and material service simultaneously preserving the required constraints remain unconnected. The conditional radiation and reciprocal absorption results are retained without identifying them directly with a first-order geometric perturbation of the original material clocks and rods.

References

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